



FRENCH BAKERY

ANALYSIS

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PROBLEM STATEMENT

OPTIMIZING SALES & INVENTORY MANAGEMENT FOR A FRENCH BAKERY

THE BAKERY IS EXPERIENCING INCONSISTENT SALES PATTERNS, LEADING TO POTENTIAL ISSUES LIKE OVERSTOCKING, UNDERSTOCKING, AND MISSED REVENUE OPPORTUNITIES. ADDITIONALLY, THE BUSINESS IS UNSURE ABOUT THE BEST-SELLING PRODUCTS, PEAK SALES HOURS, AND CUSTOMER PURCHASING BEHAVIOR, MAKING IT DIFFICULT TO OPTIMIZE PRICING, STAFFING, AND PROMOTIONS.



DATASET OVERVIEW

COLUMNS PRESENT IN DATASET :

DATE

TIME

TICKET_NUMBER

ARTICLE

QUANTITY

UNIT_PRICE



STEPS AND TOOLS

STEP 1 : IMPORTED DATA AS A DATAFRAME IN PANDAS

STEP 2 : PERFORMED EDA TO UNDERSTAND DATA

STEP 3 : DATA CLEANING

STEP 4 : FEATURE ENGINEERING

STEP 5 : ANALYSIS

**THIS PROJECT IS CREATED USING PURE PYTHON AS DATA
CONSISTS OF 200K PLUS RECORDS MAKING IT DIFFICULT TO
ANALYZE IN EXCEL DUE TO PERFORMANCE ISSUE**



Analysis

KPI'S

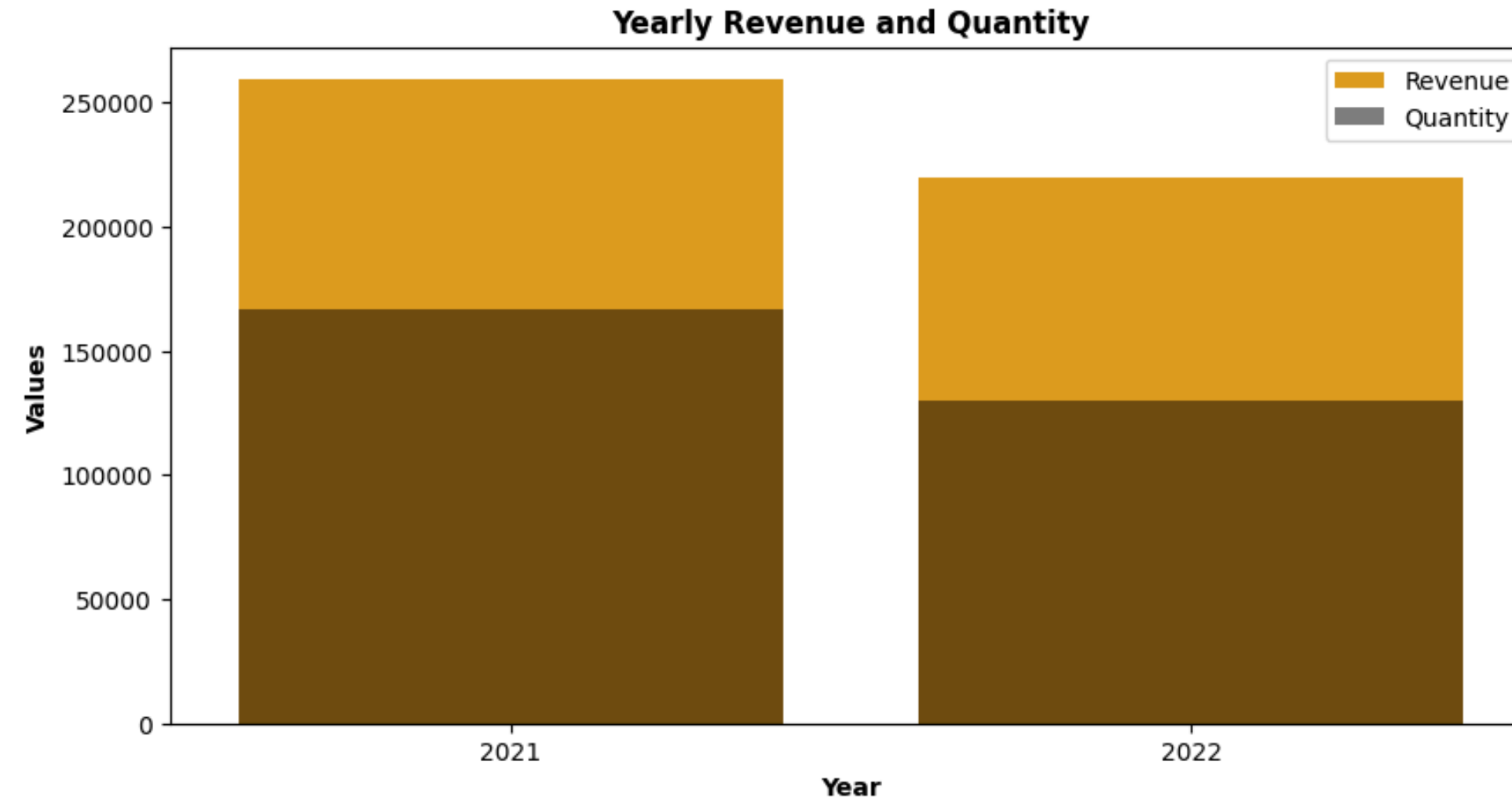
1 Total Orders : 219756

2 AVG Order Quantity : 1.349

3 Total Revenue : 478662.69

4 Average Ticket Size : 478662.69

YEARLY REVENUE AND QUANTITY

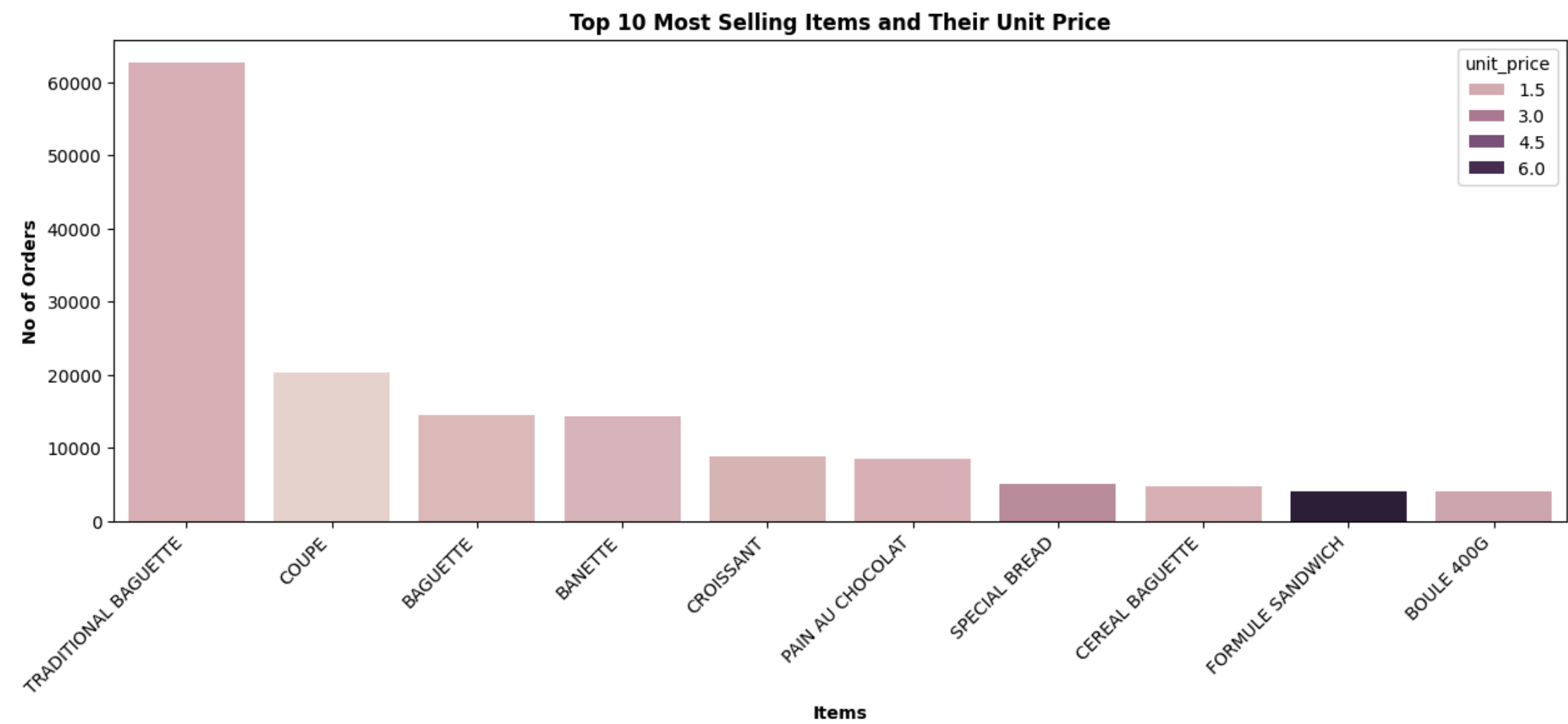


1 2021 HAD HIGHER REVENUE COMPARED TO 2022.

2 QUANTITY SOLD ALSO DECREASED IN 2022, THOUGH NOT AS SIGNIFICANTLY AS REVENUE.

3 THE GAP BETWEEN REVENUE AND QUANTITY IS LARGER IN 2021, INDICATING THAT EITHER HIGHER-PRICED PRODUCTS WERE SOLD OR PRICES WERE INCREASED IN THAT YEAR.

TOP 10 MOST SELLING ITEMS



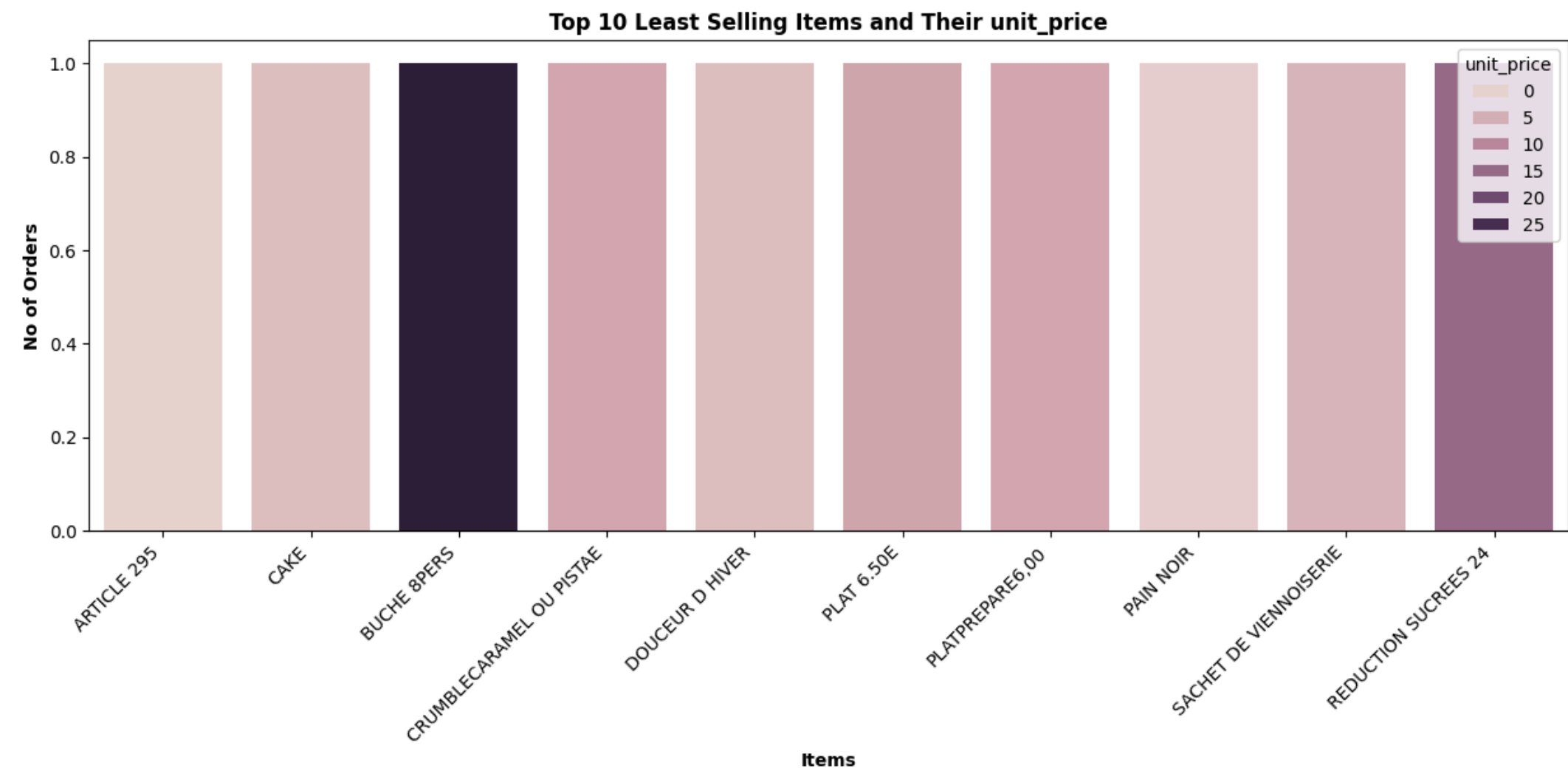
1 TRADITIONAL BAGUETTE DOMINATES SALES SIGNIFICIANTLY , OUTSPANCING OTHER ITEMS

2 OTHER POPULAR ITEMS INCLUDE COUPE, BAGUETTE, BANETTE, AND CROISSANT –INDICATING STRONG CUSTOMER DEMAND FOR BAKERY STAPLES.

3 THE UNIT PRICE IS SHADED, WITH DARKER SHADES REPRESENTING HIGHER PRICES.

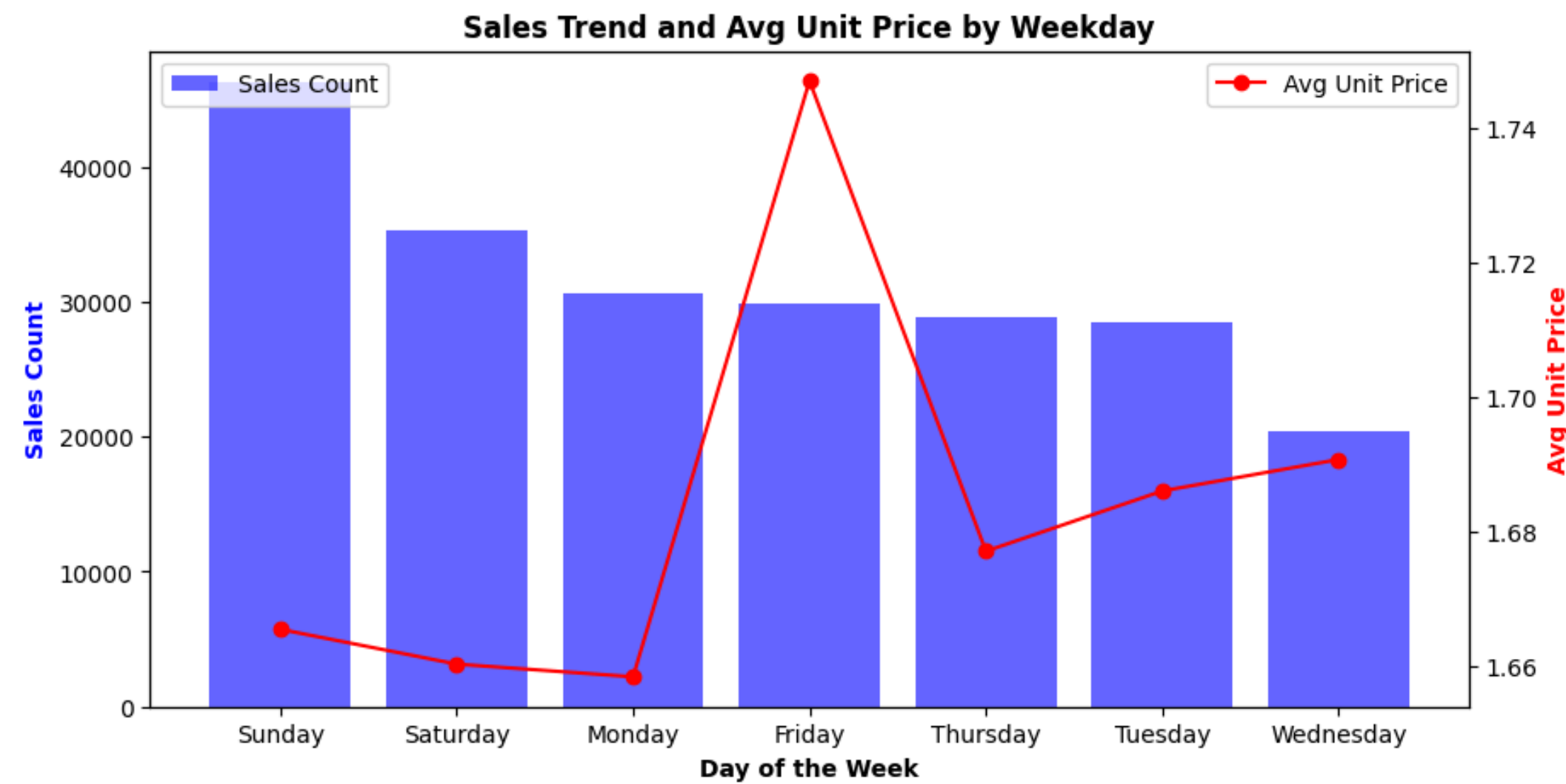
4 MOST OF THESE TOP-SELLING ITEMS SEEM TO HAVE RELATIVELY LOWER UNIT PRICES, WHICH COULD CONTRIBUTE TO THEIR HIGH SALES VOLUME.

TOP 10 LEAST SELLING ITEMS



- 1 ALL ITEMS HAVE VERY FEW ORDERS, LIKELY INDICATING LOW DEMAND.
- 2 THE UNIT PRICE IS REPRESENTED USING DIFFERENT SHADES DARKER SHADES INDICATE HIGHER PRICES.
- 3 SOME OF THESE ITEMS MAY HAVE A HIGHER UNIT PRICE, POTENTIALLY AFFECTING SALES.
- 4 CERTAIN FOOD ITEMS LIKE CAKES, TARTS, AND PREPARED PLATES APPEAR IN THE LEAST-SELLING LIST, WHICH MIGHT INDICATE CUSTOMER PREFERENCES SHIFTING AWAY FROM THESE.

SALES TREND BY WEEKDAY



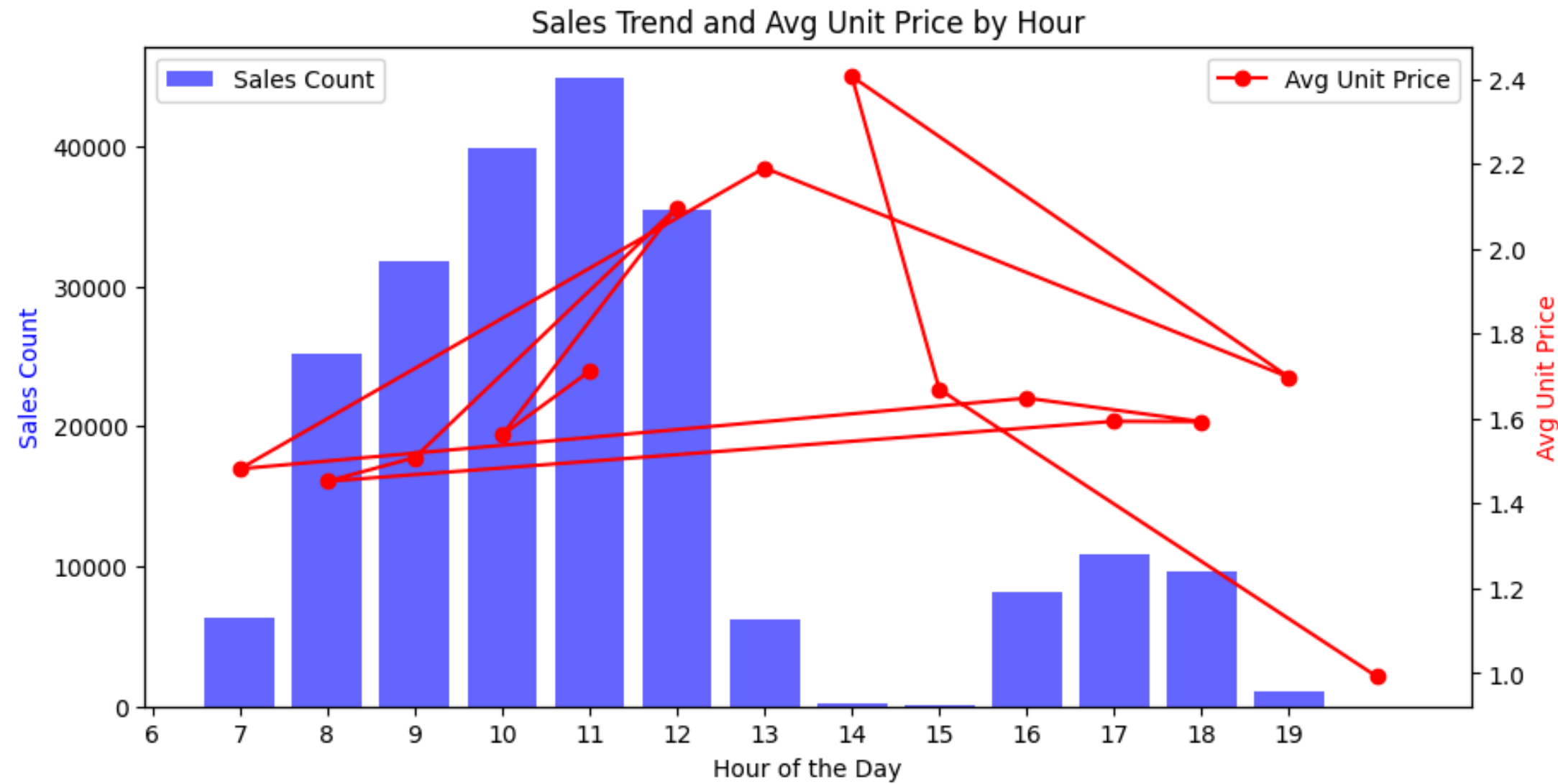
1 HIGHEST SALES : SUNDAY HAS THE HIGHEST NUMBER OF TRANSACTIONS.

2 STRONG SALES: SATURDAY, MONDAY, AND FRIDAY ALSO HAVE RELATIVELY HIGH SALES.

3 LOWER SALES : WEDNESDAY HAS THE LOWEST SALES, FOLLOWED BY TUESDAY AND THURSDAY.

4 HIGHEST AVERAGE UNIT PRICE : FRIDAY SHOWS A SIGNIFICANT SPIKE IN UNIT PRICE, WHILE SALES VOLUME IS MODERATE.

SALES TREND BY HOUR



1 HIGHEST SALES : BETWEEN 10 AM AND 12 PM, WITH THE PEAK AT 12 PM.

2 MODERATE SALES: BETWEEN 8 AM TO 9 AM AND 3 PM TO 6 PM.

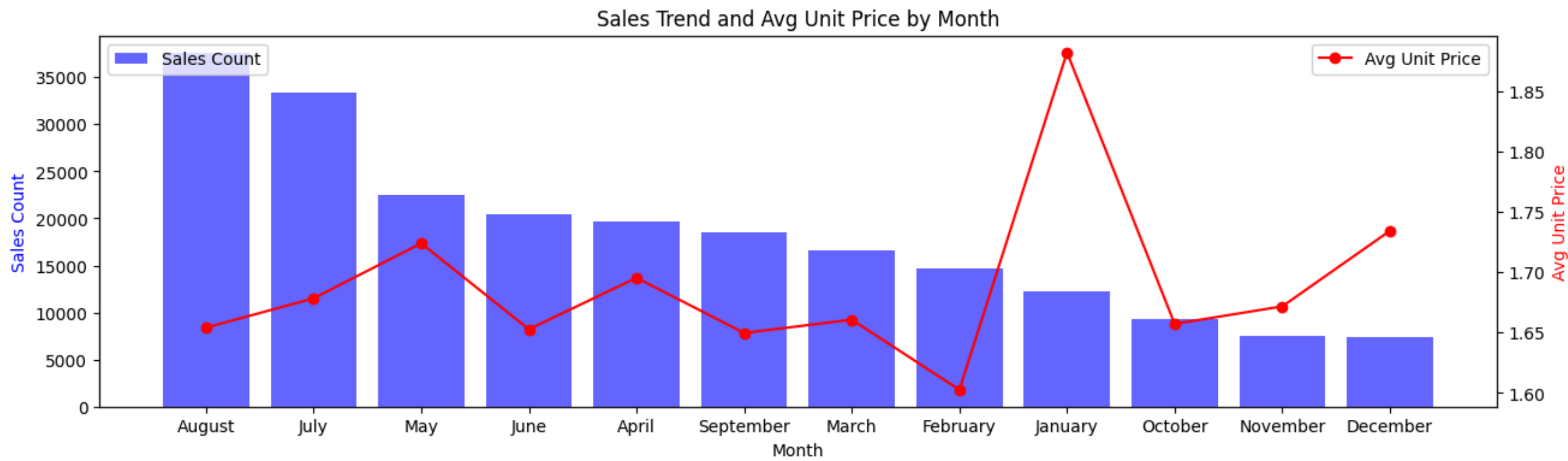
3 LOWEST SALES : (AT 7 AM AND THEN 2 , 3 , 7 PM)

4 HIGHEST AVERAGE PRICE : AROUND 2 PM, BUT IT DOES NOT CORRELATE WITH HIGH SALES.

- STABLE PRICES: BETWEEN 8 AM AND 12 PM , WHICH ALIGNS WITH PEAK SALES.

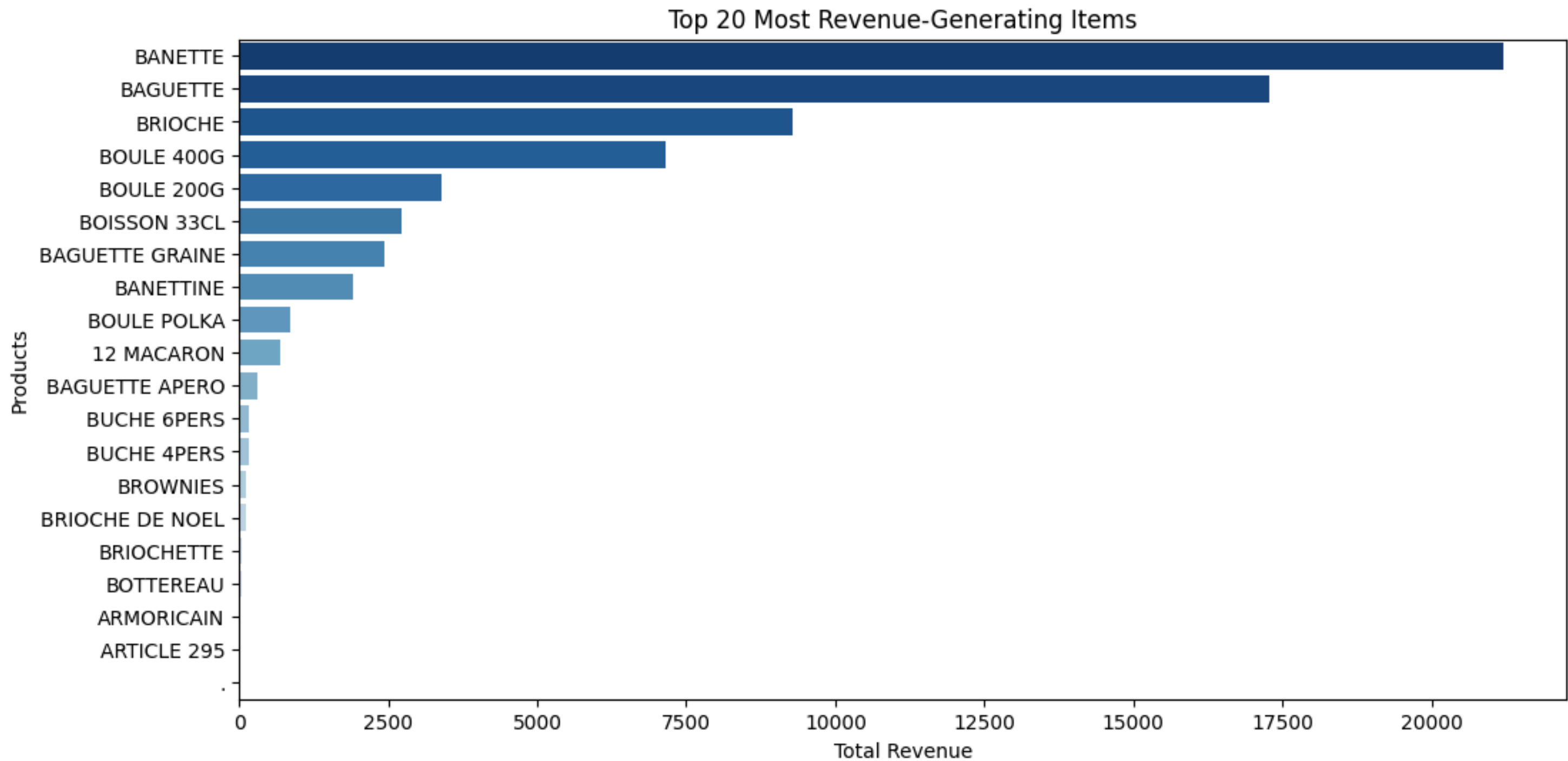
- PRICE DROP IN LATE EVENING: THE LOWEST UNIT PRICE IS AFTER 8 PM, POTENTIALLY DUE TO DISCOUNTS OR CLEARANCE SALES.

SALES TREND BY MONTH



- 1 HIGHEST SALES: AUGUST AND JULY HAD THE HIGHEST SALES COUNT.
- 2 LOWEST SALES: JANUARY AND FEBRUARY HAD THE LOWEST SALES.
- 3 MID-YEAR DROP: SALES DECLINED AFTER JULY AND REMAINED LOWER FOR THE REST OF THE YEAR.
- PRICE SURGE IN JANUARY: A SHARP INCREASE IN AVERAGE UNIT PRICE IN JANUARY.
- LOWEST PRICES IN FEBRUARY: PRICES DROPPED SIGNIFICANTLY RIGHT AFTER THE JANUARY PEAK.
- GRADUAL INCREASE TOWARD YEAR-END: PRICES SLIGHTLY INCREASED FROM OCTOBER TO DECEMBER.

TOP 20 MOST REVENUE GENERATING ITEMS



BANETTE IS THE MOST REVENUE GENERATING PRODUCTS FOLLOWED BY OTHERS

INSIGHTS AND RECOMMENDATIONS

- IN 2022 REVENUE IS LESS AS COMPARED TO 2021 SO STRATEGIES ARE REQUIRED TO ENHANCE PERFORMANCE OF BUSINESS
- BUNDLE & UPSELL: INCREASE REVENUE BY BUNDLING POPULAR BAKERY STAPLES WITH HIGHER-MARGIN OR LESS POPULAR PRODUCTS (E.G., BREAD + PREMIUM SPREADS).
- THERE IS NOT TOO STRONG BUT THERE IS SOME CORRELATION BETWEEN PRICE AND SALES SO PRICING ADJUSTMENT IS REQUIRED FOR THE ITEMS THAT ARE UNPOPULAR AND SELLING LESS .
- WEEKEND DEMAND: HIGH SALES ON SATURDAY AND SUNDAY SUGGEST THESE ARE PEAK SHOPPING DAYS. CONSIDER PROMOTIONS OR BUNDLING DEALS TO MAXIMIZE REVENUE.
- FRIDAY PRICE SURGE: THE PRICE SPIKE ON FRIDAY COULD MEAN:
 - PREMIUM-PRICED ITEMS ARE SOLD MORE.
 - PRICE HIKES BEFORE THE WEEKEND WHEN DEMAND IS EXPECTED TO INCREASE.
 - FEWER DISCOUNTS OR PROMOTIONAL OFFERS.
- MIDWEEK SALES STRATEGY: SINCE WEDNESDAY AND THURSDAY HAVE LOWER SALES, CONSIDER INTRODUCING MIDWEEK PROMOTIONS TO BOOST REVENUE.

INSIGHTS AND RECOMMENDATIONS

- **OPTIMAL PRICING STRATEGY:** PRICES ARE HIGH AROUND 2 PM, BUT SALES ARE LOW. CONSIDER ADJUSTING PRICING TO BOOST SALES IN THE AFTERNOON.
- **PROMOTION OPPORTUNITIES:** IF THE 10 AM - 12 PM SLOT IS THE BUSIEST, OFFERING BUNDLE DEALS OR UPSELLING COULD MAXIMIZE REVENUE.
- **EVENING DISCOUNT STRATEGY:** IF PRICES DROP AT NIGHT, THIS COULD INDICATE END-OF-DAY PROMOTIONS OR LOWER DEMAND.
- **SEASONALITY :** SALES PEAK IN MID-YEAR (AUGUST, JULY), POSSIBLY DUE TO SEASONAL DEMAND.
- **PRICE ELASTICITY :** HIGHER PRICES IN JANUARY DIDN'T LEAD TO HIGHER SALES.
- → SUGGESTS DEMAND IS PRICE-SENSITIVE.
- **END-OF-YEAR STRATEGY:** IF HIGHER PRICES WORK IN JANUARY BUT NOT IN OTHER MONTHS, CONSIDER PROMOTIONS OR PRICE ADJUSTMENTS.
- **DEMAND FOCUS:** BANETTE AND BAGUETTE DOMINATE SALES—ENSURE CONSISTENT STOCK AND EXPLORE PREMIUM VARIANTS.
- **UPSELLING:** BUNDLE HIGH-SELLING ITEMS WITH LOWER-PERFORMING PRODUCTS (E.G., SPREADS, COFFEE).
- **PRICING STRATEGY:** SLIGHTLY ADJUST PRICES OR INTRODUCE PREMIUM VERSIONS TO BOOST MARGINS.
- **SUPPLY CHAIN:** OPTIMIZE INVENTORY TO PREVENT STOCKOUTS AND ALIGN WITH DEMAND TRENDS.
- LET ME KNOW IF YOU NEED REFINEMENTS!

THANK YOU



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